J. Sebastian Monzon
s.monzon@yale.edu | 0000-0002-9986-4604

EDUCATION

Ph.D. in Astronomy & Astrophysics — New Haven, CT
Yale University — *Aug 2021 – present*

M.Phil in Astronomy & Astrophysics — New Haven, CT
Yale University — *Aug 2021 – May 2023*

B.S. in Astronomy — Santa Cruz, CA
University of California, Santa Cruz — *Sep 2015 - June 2019*

RESEARCH POSITIONS

Ph.D. Candidate, Yale University — July 2022 - present
Thesis Advisor: Prof. Frank van den Bosch

First-Year Research Project, Yale University — Aug 2021 - July 2022
Advisor: Prof. Marla Geha

Research Intern, NASA Ames — Feb 2020 - May 2021
PIs: Dr. Naseem Rangwala, Dr. Sarah Nickerson

Gemini/HST Cluster Project Intern, Gemini N. Observatory — May 2020 - Sep 2020
PIs: Dr. Inger Jorgensen, Dr. Kristin Chiboucas

USRA Research Assistant, NASA Ames — June 2019 - Feb 2020
PI: Dr. Randolf Klein

Undergraduate Researcher, UC Santa Cruz — June 2017 - June 2019
Advisor: Prof. J. Xavier Prochaska

PUBLICATIONS

Published (Refereed)

Monzon, J. Sebastian; van den Bosch, Frank C.; Rey, Martin P. **"Statistical Predictions of the Accreted Stellar Halos around Milky Way-Like Galaxies"**, The Open Journal of Astrophysics, 9, 160221 (2026). DOI:10.33232/001c.160221

Monzon, J. Sebastian; van den Bosch, Frank C.; Mitra, Kaustav **"Constraining the Low-mass End of the Stellar-to-halo Mass Relation with Surveys of Satellite Galaxies"**, The Astrophysical Journal, 976, 197 (2024). DOI:10.3847/1538-4357/ad834e

Nickerson, Sarah; Rangwala, Naseem; Colgan, Sean; DeWitt, Curtis; Monzon, J. Sebastian et al. **"The Mid-infrared Molecular Inventory toward Orion IRc2"**, The Astrophysical Journal, 945, 26 (2023). DOI:10.3847/1538-4357/aca6e8

Monzon, J. Sebastian; Prochaska, J. Xavier; Lee, Khee-Gan; Chisholm, John **"Effective Opacity of the Intergalactic Medium from Galaxy Spectra Analysis"**, The Astronomical Journal, 160, 37 (2020). DOI:10.3847/1538-3881/ab94c2

In Preparation

Monzon, J. Sebastian; van den Bosch, Frank C.; Mitra, Kaustav **"Halo Assembly Bias in Excursion Set-Based Subhalo Abundance Measurements"**, in preparation

Riley, Alexander H.; Shipp, Nora; Monzon, J. Sebastian; Dropulic, Adriana; Thoron, Gray; Arora, Arpit **"Simulations Agree: the Milky Way's Satellites Are Disrupting"**, in preparation for MNRAS

Nickerson, Sarah; Monzon, J. Sebastian; Rangwala, Naseem; DeWitt, Curtis; Colgan, Sean W. J. **"High Spectral Resolution Water and SiO Emission in Orion BN/KL"**, in preparation

Non-Academic

Levesque, Emily M.; Ivie, Rachel; Johns-Krull, Christopher; Krahm, Grace; Lopez, Laura A.; MacGregor, Meredith A.; Monzon, J. Sebastian et al. **"AAS Graduate Admissions Task Force: Final Report and Recommendations"**, Bulletin of the American Astronomical Society, 57(1) (2025)

Pasha, Imad; Asali, Yasmeen; Cerny, William; Monzon, J. Sebastian **"The First Mile: Bridging Introductory Programming to Astronomical Research Skills via a New Textbook, Exercise Compendium, and Seminar Framework"**, AAS Meeting Abstracts #243, 186.04 (2024)

PRESENTATIONS

Princeton "Thunch" Talk — Sep 2026
oral presentation — *Princeton, New Jersey*

Near Field Cosmology in the Era of Big Data: Local Group and Beyond — July 2026
oral presentation — *Toronto, Canada*

University of Michigan Galaxies Group Seminar Journal Club — April 2026
oral presentation — *virtual*

Rutgers University Astro Journal Club — Nov 2025
oral presentation — *Piscataway, New Jersey*

IAU Symposium 403: The Hidden Beauty of the Galactic Outskirts — Oct 2025
oral presentation — *Cordoba, Spain*

Santa Cruz Galaxy Workshop — Aug 2025
oral presentation — *Santa Cruz, California*

Galactic Frontiers II: Dwarf Galaxies in the Local Volume and Beyond — June 2025
poster presentation — *Hanover, New Hampshire*

UC Riverside AURORA Journal Club — June 2025
oral presentation — *virtual*

Harvard ITC Luncheon — Feb 2025
oral presentation — *Cambridge, Massachusetts*

Santa Cruz Galaxy Workshop — Aug 2024
oral presentation — *Santa Cruz, California*

Dwarf Galaxies, Star Clusters and Stellar Streams — July 2024
poster presentation — *Chicago, Illinois*

Galactic Frontiers: Dwarf Galaxies in the Local Volume and Beyond — July 2023
oral presentation — *New York, New York*

241st AAS Conference — Jan 2023
poster presentation — *Seattle, Washington*

NASA Ames Center-Wide Internship Presentations — Aug 2020
oral presentation — *Mountain View, California*

235th AAS Conference — Jan 2020
poster presentation — *Honolulu, Hawaii*

Revealing Cosmology and Reionization History with the IGM — Sep 2018
poster presentation — *Tokyo, Japan*

SERVICE & LEADERSHIP

AAS Committee for the Status of Minority Students in Astronomy (CSMA) — Jan 2024 - present

- Published recommendations aimed at improving the state of graduate admissions in astronomy
- Serving on a graduate admissions task force
- Participating in monthly committee meetings

Astronomy Graduate Congress (AGC) — Sep 2023 - Dec 2024

- Co-created a new organization of astronomy graduate students across the country
- Developed student advocacy cheat-sheet
- Co-organized Speak Out for Science - Testimonial Day of Action with the AAS
- Developing projects aimed at increasing a sense of community and equity in graduate programs

Yale's Astronomy Climate and Diversity Committee — Sep 2022 - present

- Developing undergraduate workshops to maximize effective access to research

- Advising faculty about equitable practices concerning qualifying exams
- Overseeing organizational tasks to keep the committee active and efficient

Yale SACNAS Chapter Dec 2023 - April 2025

- SACNAS: Society for Advancement of Chicanos/Hispanics and Native Americans in Science
- Chaired the professional development committee
- Facilitated the 2025 SACNAS New England Chapters Community Gathering
- Participating in monthly organizational meetings

Graduate Student Assembly Sep 2023 - June 2024

- Participating in bi-weekly assemblies as the elected department delegate
- Advising and reporting on issues concerning graduate student wages
- Acting as channel of communication between faculty and graduate students

Awards and Recognition

- Diversity, Equity, Inclusion & Belonging Departmental Service Award (recognition for outstanding departmental service) 2024
- Yale Teaching Innovation Project Grant (university support for AstroRPS) 2023
- Yale Gruber Science Fellow (awarded to the most highly ranked incoming Yale PhD applicants) 2021
- UC Santa Cruz Senior Thesis Honors, Dean's Honors (recognition for academic excellence) 2019
- Lamat REU Fellowship (undergraduate research fellowship) 2017, 2018

Teaching & Mentoring

- Undergraduate (Senior Thesis) Research Project Mentor Sep 2026 - present
- Professional Development Workshops for Central American - Caribbean Bridge in Astrophysics Sep 2025 - present
- Yale Astronomy Research Preparation Seminar (Astro-RPS) Sep 2024 - present
- Yale Graduate Student Teaching Assistant Aug 2021 - May 2024
- Yale AstroSibs Oct 2021 - present
- Mini Research Project for YSPA High School Student Aug 2023 - Dec 2023
- UCSC Academic Excellence Program Co-leader Aug 2018 - June 2019

Outreach

- Science in the News (public science communication at local libraries) 2025
- Spanish Night at Leitner Family Observatory (Spanish-language astronomy outreach at local planetarium) 2024
- Astro on Tap (public astronomy talks and engagement) 2022, 2023, 2024